

Training Course

Amazon Web Service



Module 11: CI/CD With AWS



■ **Goal:** Understanding CI/CD Services in AWS

- ✓ AWS CodePipeline
- ✓ AWS CodeCommit
- ✓ AWS CodeBuild
- ✓ AWS CodeDeploy
- ✓ Deployment Strategies
- ✓ **Prerequisites:**
- ✓ CI/CD And Devops Related Knowledge
- ✓ Containerization

Lab: Create and configure CodePipeline Deploy on ECS

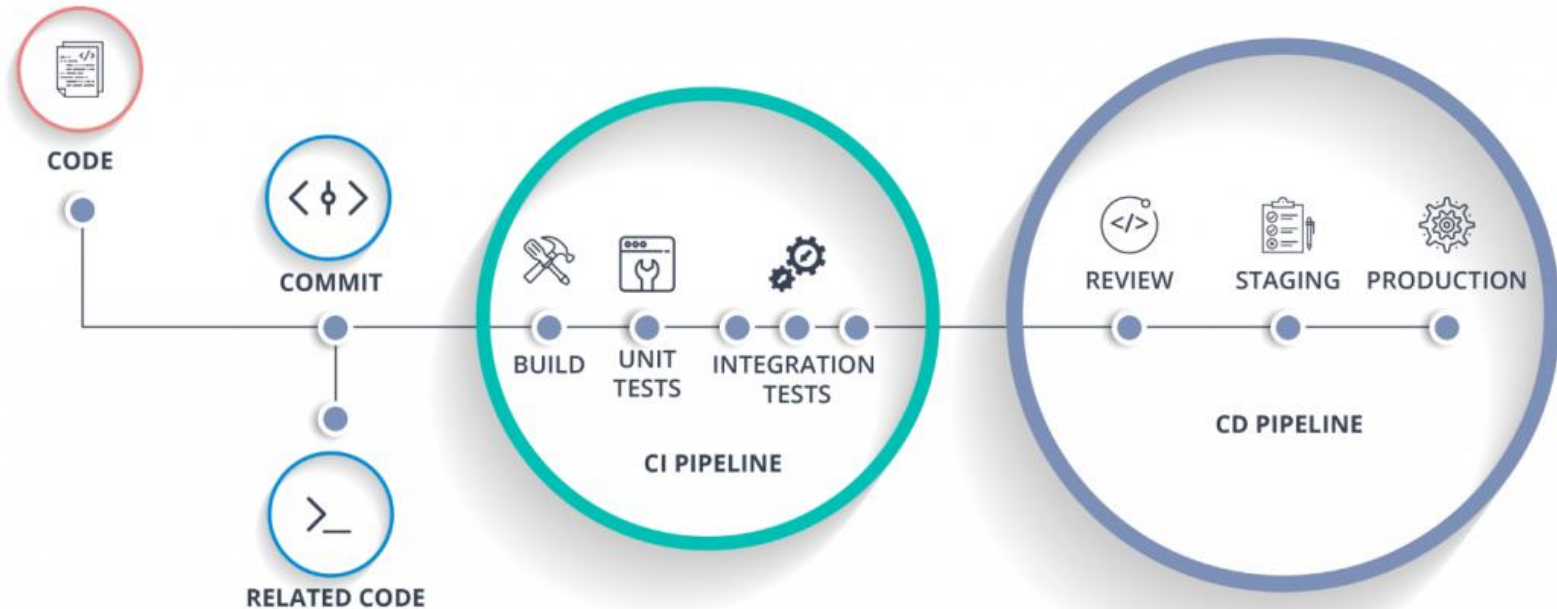
Introduction

- CI/CD stands for Continuous Integration and Continuous Deployment. It is a software development practice that involves automatically building, testing, and deploying code changes to various environments.
- Continuous Integration (CI) refers to the practice of merging code changes into a shared repository multiple times a day. Automated testing is run on the code changes to ensure they do not break the build.
- Continuous Deployment (CD) refers to the practice of automatically deploying code changes to various environments such as development, staging and production.

Introduction

- The goal of CI/CD is to catch and fix errors early in the development process, reduce lead time and provide faster feedback to the development team.
- Implementing CI/CD can improve the quality of the code and reduce the risk of introducing bugs, improve the collaboration and communication between developers and reduce the time it takes to deliver new features.

Introduction



Introduction

- Code SVM: AWS CodeCommit
- CI: AWS CodeBuild
- CD: AWS CodeDeploy
- Artifact Storage: AWS CodeArtifact
- Orchestrate Pipeline: AWS CodePipeline

AWS CodeCommit

- CodeCommit is a fully-managed source control service that makes it easy for companies to host their Git repositories in the cloud.
- With CodeCommit, you can store your source code and other assets in a private, secure, and highly available repository.
- CodeCommit supports all Git features such as branching, merging, and pull requests.
- You can access the repositories from anywhere and collaborate with your team using IAM permissions.

AWS CodeCommit

- CodeCommit also integrates with other AWS services such as CodeBuild, CodeDeploy, and CodePipeline to provide a seamless CI/CD experience.
- To create a new repository on CodeCommit, you can use the AWS Management Console, the AWS CLI, or the CodeCommit API.
- You can also import existing Git repositories from other sources such as GitHub, Bitbucket, or a local file system.
- CodeCommit also provides a number of security features such as encryption at rest, integration with IAM and KMS, and the ability to audit repository access using CloudTrail.

AWS CodeCommit

The screenshot displays the AWS CodeCommit console interface. On the left, a sidebar menu under 'Developer Tools' lists various options, with 'Commits' highlighted. The main area shows the 'MyDemoRepo' repository with the 'Commit visualizer' tab selected. This tab displays a commit history graph and a list of recent commits.

Developer Tools **CodeCommit**

▼ Source • CodeCommit

- Getting started
- Repositories
- Code
- Pull requests
- Commits**
- Branches
- Tags
- Settings

► Build • CodeBuild

► Deploy • CodeDeploy

► Pipeline • CodePipeline

Developer Tools > CodeCommit > Repositories > MyDemoRepo > Commits

MyDemoRepo

Commits | **Commit visualizer** | Compare commits

Commit visualizer

The commit visualizer shows a graph of commit history with a vertical timeline and colored lines representing branches. The following table lists the commits shown in the interface:

Commit ID	Commit Message	Time Ago
d615e7ae	Merge branch 'AnotherBranch' into testbranch	2 minutes ago
b6589863	Added another file.	2 minutes ago
73a6e39c	remote-tracking branch 'refs/remotes/origin/jane-branch' into jane-branch	
6bbb6d3c	Another test of the editing feature.	20 minutes ago
edacdf8e	Testing this out to see how well it works.	23 minutes ago
70bb94d7	Revised test results with correct information.	36 minutes ago
b78e6d1c	Merge branch 'results' into testbranch	50 minutes ago
84b7d158	Edited ahs_count.py	50 minutes ago

AWS CodeBuild

- CodeBuild is a fully-managed build service that compiles source code, runs tests, and produces software packages that are ready for deployment.
- CodeBuild allows you to build various types of applications such as Java, .NET, Ruby, Go, and more. It also supports building container images using Docker.
- CodeBuild can be configured to run on a schedule, in response to a code change, or as part of a pipeline in CodePipeline.
- CodeBuild uses build environments, which are pre-configured with specific versions of programming languages, tools, and frameworks.

AWS CodeBuild

- You can also create custom build environments to meet specific requirements.
- CodeBuild integrates with other AWS services such as CodeCommit, CodeDeploy, and CodePipeline to provide a seamless CI/CD experience.
- CodeBuild also provides detailed logs, metrics, and notifications to help you troubleshoot and optimize your builds.
- For security, CodeBuild integrates with IAM, so you can control access to the build environments and the artifacts produced by the builds.

AWS CodeBuild

- `buildspec.yaml` file:
 - Can place in the code repository or in the configuration of that code build.
 - This is a yaml file (the purpose of this file is similar to jenkins file)
 - This file is used for instructing CodeBuild on how to build the project (using script code)

AWS CodeBuild

buildspec.yml x

```
1  version: 0.2
2
3  env:
4    variables:
5      JAVA_HOME: "/opt/jdk1.8.0_131"
6      JRE_HOME: "/opt/jdk1.8.0_131/jre"
7      PATH: "$PATH:/opt/jdk1.8.0_131/bin:/opt/jdk1.8.0_131/jre/bin"
8      PATH: "$PATH:/opt/apache-maven-3.5.0/bin"
9  phases:
10   install:
11     commands:
12       - apt-get update
13       - apt-get install maven
14       - apt-get install wget
15       - apt-get install tar
16
17   pre_build:
18     commands:
19       - wget http://mirror.fibergrid.in/apache/maven/maven-3.5.0/binaries/apache-maven-3.5.0-bin.tar.gz -P /opt/
20       - tar -xzf /opt/apache-maven-3.5.0.tar.gz
21
22   build:
23     commands:
24       - mvn install -DskipTests=true
```

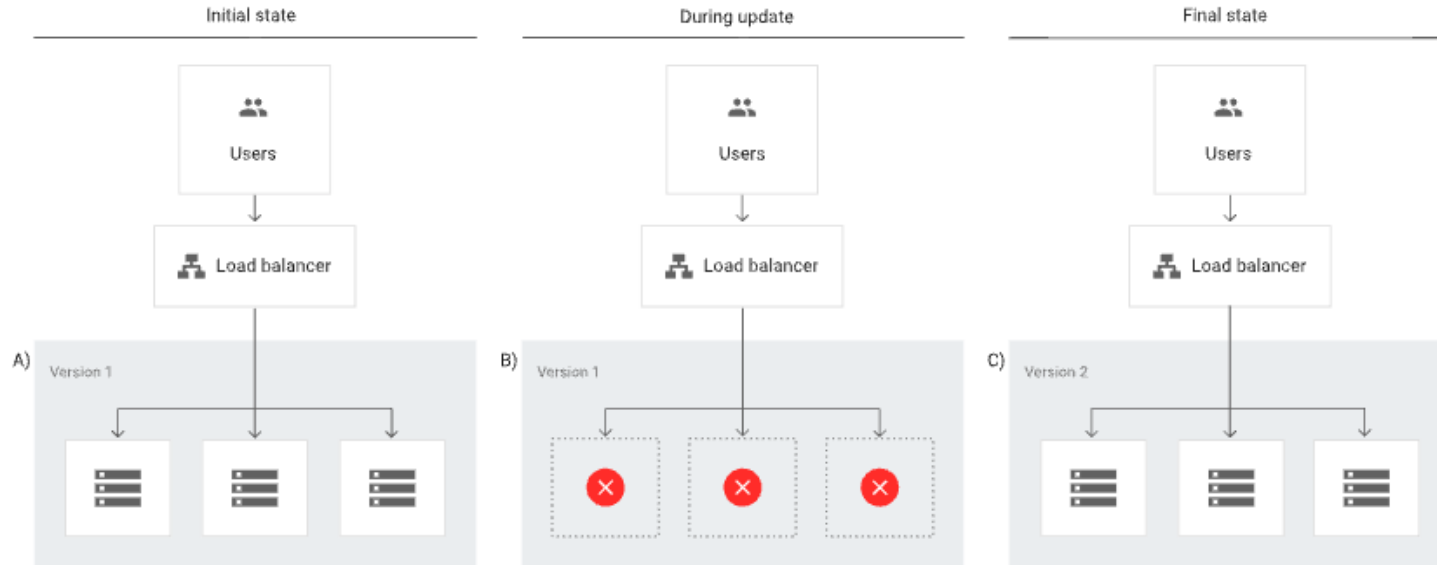
buildspec.yaml
file example

Deployment Strategy

Strategy	ZERO DOWNTIME	REAL TRAFFIC TESTING	TARGETED USERS	CLOUD COST	ROLLBACK DURATION	NEGATIVE IMPACT ON USER	COMPLEXITY OF SETUP
RECREATE version A is terminated then version B is rolled out	✗	✗	✗	■□□	■■■	■■■	□□□
RAMPED version B is slowly rolled out and replacing version A	✓	✗	✗	■□□	■■■	■□□	■□□
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■■■	□□□	■■■	■■■
CANARY version B is released to a subset of users, then proceed to a full rollout	✓	✓	✗	■□□	■□□	■□□	■■■
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■□□	■□□	■□□	■■■
SHADOW version B receives real world traffic alongside version A and doesn't impact the response	✓	✓	✗	■■■	□□□	□□□	■■■

Deployment Strategy

- Recreate:



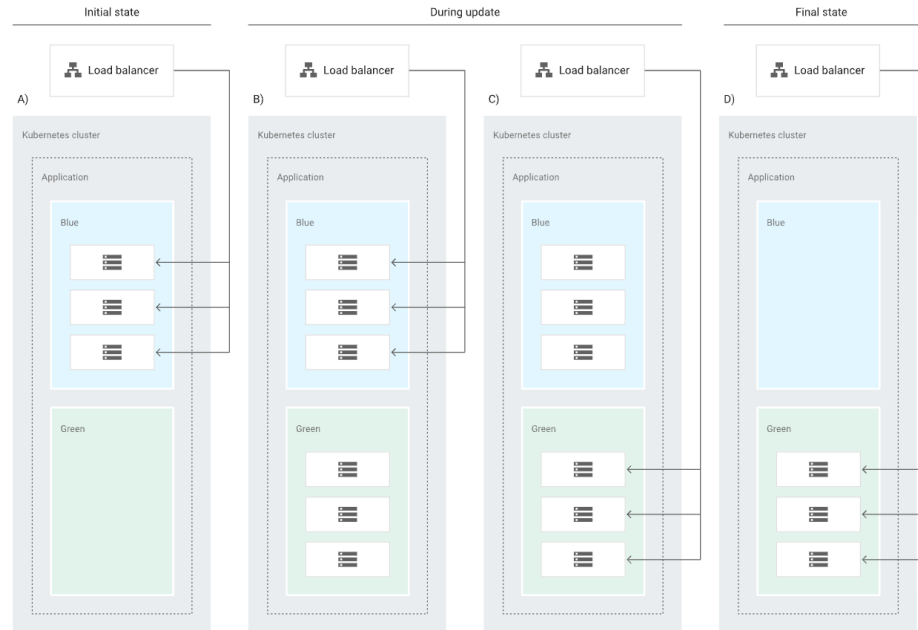
Deployment Strategy

- Ramped/Rolling Update:



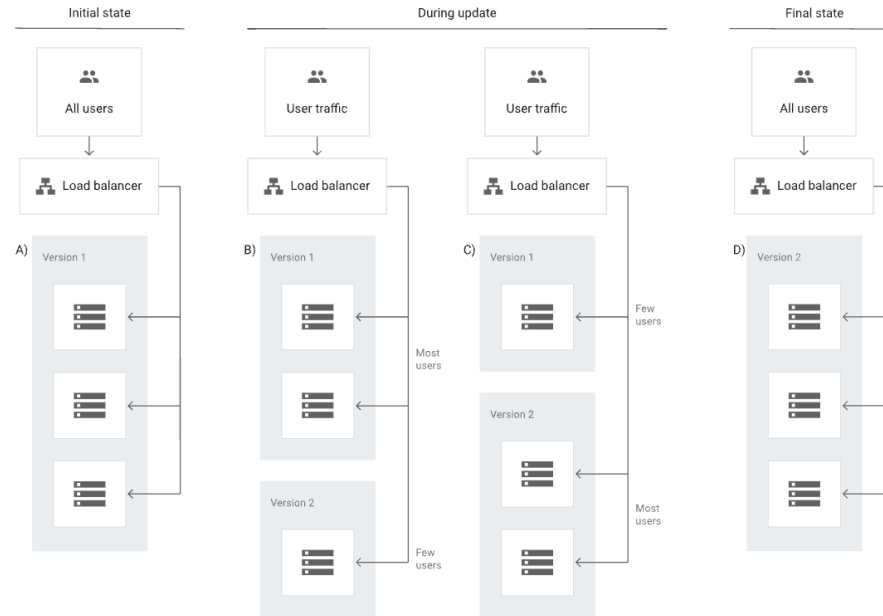
Deployment Strategy

- Blue/Green:



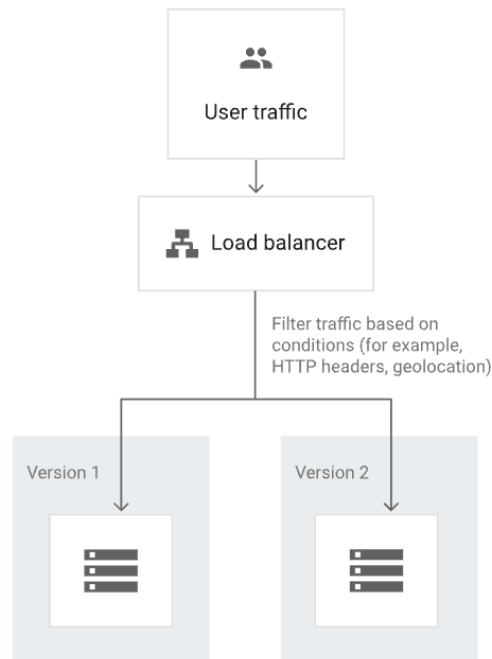
Deployment Strategy

- Canary:



Deployment Strategy

- A/B Testing:



A subset of user is
routed to the version 2

AWS CodeDeploy

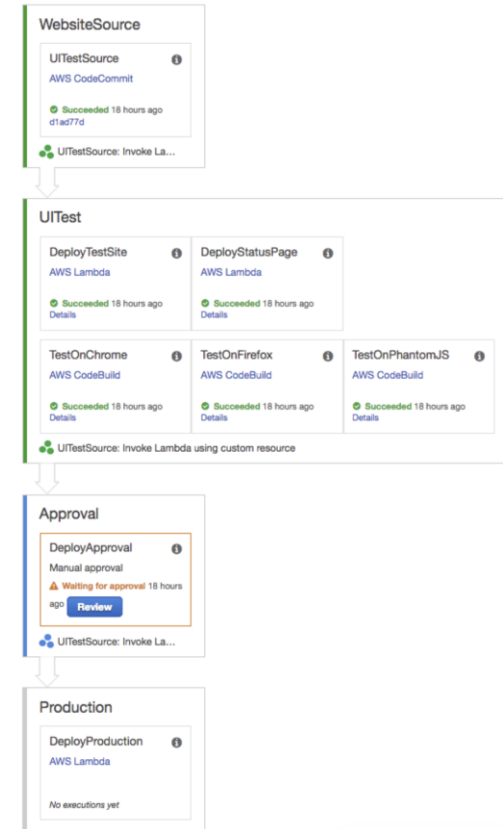
- CodeDeploy is a fully-managed deployment service that automates software deployments to a variety of compute services such as Amazon EC2, AWS Fargate, AWS Lambda, and on-premises servers.
- CodeDeploy can be used to deploy new versions of an application or to update the configuration of an existing application.
- CodeDeploy supports both in-place and blue/green deployment strategies, which allows for minimal downtime during deployment.

AWS CodeDeploy

- CodeDeploy also integrates with other AWS services such as CodeCommit, CodeBuild, and CodePipeline to provide a seamless CI/CD experience.
- CodeDeploy also provides detailed logs, metrics, and notifications to help you troubleshoot and optimize your deployments.
- CodeDeploy also integrates with IAM, so you can control access to the deployment groups and the instances in the deployment group.

AWS CodePipeline

- CodePipeline is a fully-managed continuous delivery service that helps automate the release pipeline for fast and reliable application and infrastructure updates.
- CodePipeline enables you to design and visualize your release pipeline, and automate the steps required to release your software.



AWS CodePipeline

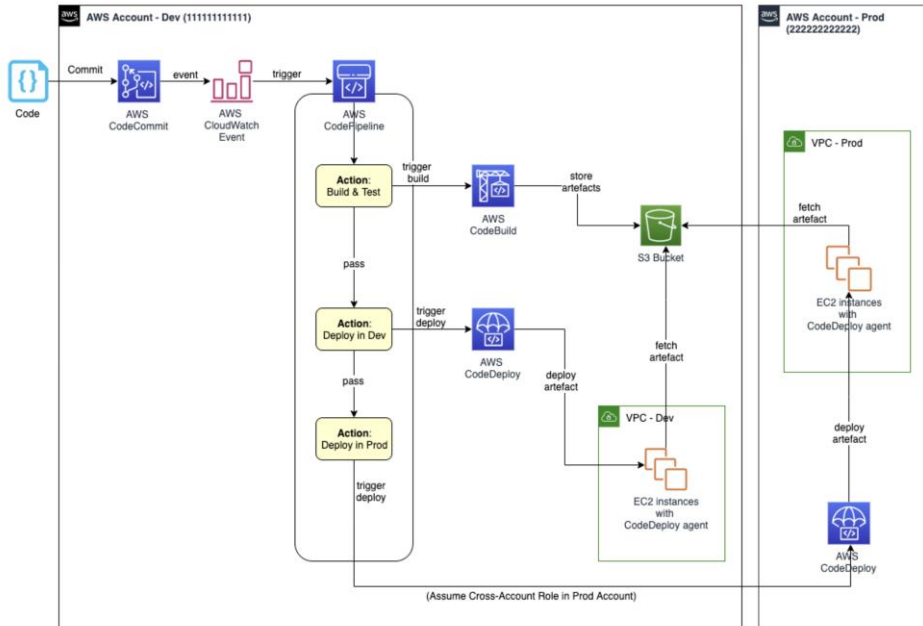
- CodePipeline allows you to integrate with other AWS services such as CodeCommit, CodeBuild, and CodeDeploy, as well as third-party services like GitHub and Jenkins.
- CodePipeline supports multiple stages and actions, such as building, testing, and deploying your code.

AWS CodePipeline

- CodePipeline also supports automatic rollbacks in case of a failure during deployment
- CodePipeline also provides detailed logs, metrics, and notifications to help you troubleshoot and optimize your pipelines.
- CodePipeline also integrates with IAM, so you can control access to the pipeline, stages, and actions.

AWS CodePipeline

- CodePipeline example design architecture:



Note: CodePipeline can deploy cross-account (more advanced with lots of configuration needed)

Thank you!!!